Hugo Marquerie 22/01/2026

Lem aut disco unidad composicion tau

Lema 1. *Sean $a, b \in \mathbb{D}$, entonces*

$$\tau_b \circ \tau_a = \rho_\gamma \circ \tau_c \quad \textit{donde} \quad c = \tau_a(b) = \frac{a-b}{1-\overline{a}b} \quad \wedge \quad \gamma = \begin{cases} \tau_{\overline{a}b}(1) & \textit{si } a \neq b, \\ -1 & \textit{si } a = b. \end{cases}$$

***Demostración*:** Por el `Teo-aut-disco-unidad-parametros/`**??**, sabemos que

$$c = (\tau_b \circ \tau_a)^{-1}(0) = \left(\tau_a^{-1} \circ \tau_b^{-1}\right)(0) = \tau_a(\tau_b(0)) = \tau_a(b) = \frac{a-b}{1-\overline{a}b}.$$

Además, tenemos que $(\tau_b \circ \tau_a)(0) = \rho_\gamma(c)$, luego $\tau_b(a) = \gamma\, \tau_a(b)$ y como $a \neq b$, $\tau_a(b) \neq 0$, por lo que

$$\gamma = \frac{\tau_b(a)}{\tau_a(b)} = \frac{{(b-a)}\big/{(1-\bar{b}a)}}{{(a-b)}\big/{(1-\overline{a}b)}} = \frac{\overline{a}b-1}{1-a\overline{b}} = \tau_{\overline{a}b}(1).$$

■

Referenciado en

- `teo-aut-disco-unidad-composicion`

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